USB Wireless Clients

Notebooks are WiFi-enabled using PC cards while desktop PCs use PCI WiFi cards to do so. Another solution is to use an external USB wireless adapter. These support both 802.11g as well as 802.11b, and can be used with notebooks and desktop PCs as well.

Those supporting 802.11b come with USB 1.1, while the ones supporting 802.11g are equipped with USB 2.0. The USB interface itself does not impose any restriction on data transfer rates. All the three tested devices put up performances as good as any other PCI or PC wireless card.

If your system does not have a free PCI slot, or if the laptop's PCMCIA slot is already in use, you have no option but to buy a USB-based wireless client.

These devices are costlier than the PCI/PCMCIA option, but do remember that an external USB device can provide you with the option of enabling WiFi functionality on your home as well as office PC.

Data Throughput (MPS)	Dax DX-900USB (802.11b)	Compex WLU54G (802.11g)	Compex WLU11A (802.11b)
Compex 26G	.39	.52	.37
Linksys WPC11	.13	.13	.17
D-Link DWL-2000AP+	.36	.62	.33
Latency (ms)			
Compex 26G	2	2	82
Linksys WPC11	25	5	4
D-Link DWL-2000AP+	3	3	3
SNR (db)			
Compex 26G	100	-51	NC
Linksys WPC11	100	-55	NC
D-Link DWL-2000AP+ (db)	100	-60	NC
Price (Rs)	3815	4400	2600

Installing PCMCIA cards is not much of a hassle

SMC 2802W

here, ESSID is optional—it just the name by which the AP will be recognised. The channel should be specified so that there is no overlapping. Next, you will need to specify WAN details, which will be provided by your ISP. Finally, you need to set up security. Here, give preference to WPA—it is more secure.

Making your PCs Wi-Fi-enabled

APs, by themselves, do not make a wire-free zone; you need to purchase PCI cards for your computers. Just like the APs, PCI cards also come in 802.11g and 802.11b flavours. In this test, we saw five cards supporting 802.11g, while only one represented the 802.11b category. This clearly indicates that the market is moving towards 802.11g.

AirPlus G+ DWL-520G+

PCI CARDS@802.11G

Performance

In this test, Buffalo's PCI card achieved a maximum data rate of 1.17 MBps when connected to a Compex 26G AP, but the pair was not able to give sustained performance. NetGear's WG511, though a little slower, showed a sustained data throughput across APs and across locations, thus being the performer of interest. The best two turned out to be NetGear's WG311 and Buffalo's WLI-PCI-G54.

Features

PCI cards with antennas that are flexible wires about two metres long help you get better connectivity than those that provide a fixed antenna that protrudes from behind the cabinet. These tend to get blocked in corners, and thus cause performance to degrade. Buffalo and MSI provide antennas with flexible wires.

Unlike PC cards, all PCI cards provide an antenna. Buffalo's antenna connector, though, is not standard: you cannot connect a replacement antenna if the original gets damaged. Also missing in the Buffalo device was WPA.

Ease Of Use

There is not much to configure in a wireless client—just install the drivers and connect it to the AP. These steps are essentially the same, irrespective of the card you are using. Still, a quick-start guide and a configuration utility can come in handy.

Price

Here again, the performers were not the winners: pricing causes both Buffalo's WLI-PCI-G54 and NetGear's WG311 to lose out to SMC's 2802W and D-Link's AirPlus G+ DWL-G520+.

Adding Hardware

For PCI cards, there is not much of a hardware installation hassle. Find a free PCI slot on the motherboard and insert the wireless card. Connect the external antenna, if one comes bundled. Remember to switch off the power supply before doing a surgery on your PC's internals.

Driver Installation

On restarting the PC after you insert the PCI card, Windows will ask for the driver disk. Insert the driver CD-ROM that comes with the card. You can switch on the APs now. As the driver installation completes itself, you should see a network icon in the system tray. Right-click on this icon and select 'View Available Wireless Networks'. Now, you should be able to see the AP. Select it, then

The Standards Battle

	802.11a	802.11g	802.11b
Data Rate (Mbps)	54	54	11
Avg. Actual Data Throughput (Mbps)	25 - 30	20 - 25	4 -5
RF Band (GHz)	5	2.4	2.4
Channels	12	11	11
Non-overlapping channels	8	3	3